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SEMI E116-0324
SPECIFICATION FOR EQUIPMENT PERFORMANCE TRACKING

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NOTICE: The section symbol (§) is used to reference a numbered section, paragraph, or header and all subordinate paragraphs, headers, and embedded material (i.e., EXCEPTIONs, lists) therein. The paragraph symbol (¶) is used to reference a particular numbered paragraph and embedded material therein. When duplicated (i.e., §§ and ¶¶) the symbols are used to reference more than one section or paragraph, respectively.

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1 Purpose

1.1 This Standard provides specification for implementing basic equipment performance tracking (EPT) for production equipment. Provisions in this Standard enable the host computer to track basic equipment performance in an automated and consistent matter, without operator or host input. This Standard provides specification for equipment suppliers to:

- Track basic equipment states (no operator or host input required).
- Track basic equipment states in a modular manner, for both major modules and the overall equipment.
- Report basic equipment state changes to a host computer, at both module and equipment level.
- Report equipment's time in state to a host computer, at both module and equipment level.
- Report reasons to a host computer for why equipment is blocked from performing its task, at both module and equipment level.

1.2 Equipment users require the ability to track equipment performance without dependence on user input to eliminate inaccuracies due to incorrect or untimely input from the user. The ability to track equipment performance without user input is essential in 300 mm wafer factories where operational scenarios require minimal manual interaction. EPT defines concepts, behavior, and message services that enable the host computer to obtain the equipment data required for equipment performance tracking in an automated and consistent matter without operator or host input.

1.3 EPT enables factory managers to identify the current states of factory equipment, both at the equipment level and at the module level (e.g., processing chambers), without dependence on user input. EPT enables factory engineers to evaluate the time that equipment and modules spent in different states and identify areas for improvement. EPT enables factory engineers to obtain directly from the equipment the reasons why the equipment or module is prevented from performing. EPT provides industrial engineers the equipment data which, when combined with external data from the Manufacturing Execution System (MES), will enable accurate calculation of SEMI E10 states and SEMI E79 metrics at the equipment-level and the module-level. EPT enables automation engineers to develop reusable host interfaces by using a standardized collection event and data variables to collect equipment state data.

2 Scope

2.1 The Scope of this Standard is to define equipment behavior states and the data required to track basic equipment performance for production equipment. These requirements are intended to facilitate equipment-level and module-level state tracking and to communicate state information to the host for simple equipment performance tracking, without requiring host or operator input.

2.2 Specifically, this Standard provides:

- An EPT state model that defines triggers for state changes.
- Specification of data variables required to communicate basic equipment performance data to the host computer.

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- Specification of event messages used to communicate basic equipment performance data to the host computer.
- Requirements for EPT compliance.

2.3 This Standard specifies the concepts, behavior, and message services that enable the host computer to obtain the equipment data required for equipment performance tracking in an automated and consistent manner, without operator or host input. It does not specify the report of SEMI E10 states from the equipment to a host computer, as this information requires user input and is already specified by SEMI E58.

2.4 This Standard does not conflict with SEMI E58 nor does it inhibit the equipment's ability to report out SEMI E10 states via SEMI E58.

2.5 This Standard is a building block towards SEMI E10 and SEMI E79 by providing accurate equipment information required for SEMI E10 and SEMI E79 metrics, without dependence on operator or host input, eliminating inaccuracies due to incorrect or untimely input from the user. EPT assists both SEMI E10 and SEMI E79 by providing a modular approach to equipment performance tracking, allowing the state of the equipment to be determined by the states of its major modules. EPT assists SEMI E79 by providing task-level detail of the equipment or module's current activity, allowing performance metrics to be tracked at the task level.

NOTICE: SEMI Standards and Safety Guidelines do not purport to address all safety issues associated with their use. It is the responsibility of the users of the Documents to establish appropriate safety and health practices, and determine the applicability of regulatory or other limitations prior to use.

3 Limitations

3.1 This Standard applies to semiconductor production equipment. Other types of equipment have not been examined. However, it may be used for other types of equipment when applicable.

4 Referenced Standards and Documents

4.1 *SEMI Standards and Safety Guidelines*

SEMI E5 — Specification for SEMI Equipment Communications Standard 2 Message Content (SECS-II)

SEMI E10 — Specification for Definition and Measurement of Equipment Reliability, Availability, and Maintainability (RAM) and Utilization

SEMI E30 — Specification for the Generic Model for Communications and Control of Manufacturing Equipment (GEM)

SEMI E39 — Specification for Object Services: Concepts, Behavior, and Services

SEMI E58 — Automated Reliability, Availability, and Maintainability Standard (ARAMS): Concepts, Behavior, and Services

SEMI E79 — Specification for Definition and Measurement of Equipment Productivity

SEMI E172 — Specification for SECS Equipment Data Dictionary (SEDD)

NOTICE: Unless otherwise indicated, all documents cited shall be the latest published versions.

5 Terminology

5.1 *Abbreviations and Acronyms*

5.1.1 *EFEM* — equipment front end module

5.1.2 *EPT* — equipment performance tracking

5.2 *Definitions*

5.2.1 *EPT module* — a major component of the equipment that affects processing or throughput.

5.2.1.1 For purposes of simplification, an EPT module executes one and only one task at a time. Each EPT Module has an EPT state model that is maintained by the equipment.

5.2.2 *EPT state* — the state of IDLE, BUSY, or BLOCKED within EPT state model.

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5.2.3 *event report* — a message the equipment sends to the host on the occurrence of a collection event.

5.2.4 *fault* — an exception.

5.2.5 *host* — the factory computer system or an intermediate system that represents the factory and the operator to the equipment.

5.2.6 *intended function* — a manufacturing function that the equipment was built to perform. This includes transport functions for transport equipment, measurement functions for metrology equipment, as well as process functions such as physical vapor deposition and wire bonding. Complete equipment may have more than one intended function. [SEMI E58]

5.2.7 *material* — the basic unit of process. For the purposes of this Standard, material is a set of one or more substrates.

5.2.8 *module* — a major component of equipment that contains at least one material location and performs some task on material. Equipment modules may be aggregates of equipment subsystems, I/O devices, and other modules. [SEMI E98]

5.2.9 *substrate* — the basic unit of material on which work is performed to create a product. Examples include wafers, die, plates used for masks, flat panels, circuit boards, and leadframes. [SEMI E90]

5.2.10 *task* — a planned and repeatable activity with an expected duration and a definite beginning and end (e.g., Move wafer from cassette to stage, Pre-align wafer, Align reticle, Preheat chamber, Increase vacuum). NOTE: Actual durations may vary.

5.2.11 *trigger* — an event that causes a change in the state of the equipment. Examples are changes in sensor readings, alarms, and messages received from the host, and operator commands.

5.2.12 *unit* — any wafer, die, packaged device, or piece thereof (included product and non-product units). [SEMI E10]

5.2.13 *user* — any entity interacting with the equipment, either locally as an operator or remotely via the host. From the equipment's viewpoint, both the operator and the host represent the user. [SEMI E58]

6 Requirements

6.1 An EPT compliant implementation requires provision of certain capabilities defined by other standards: accessibility to status information, event reporting, alarm management, and provision of an internal time-and-date clock. These requirements may be satisfied through compliance to SEMI E30 for the following sets of requirements:

- Clock Services
- Event Notification
- Status Data Collection
- Equipment Constants
- Alarm Management

6.2 An EPT compliant implementation requires a documented list of all EPT modules contained within the equipment.

6.3 An EPT compliant implementation requires an EPT state model for the equipment and for each EPT module.

6.4 Each EPT module shall have a documented list of tasks that the module may perform.

7 Conventions

7.1 This Standard follows the conventions for state model methodology and service definitions used by the SEMI Standards referenced in § 4.

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7.2 *State Model Methodology*

7.2.1 A state model has three elements: definitions of each state and substate, a diagram of the states showing the valid transitions between states, and a corresponding state transition table that defines the triggers for each transition. The diagram of the state model uses the Harel State Chart notation. An overview of this notation is presented in SEMI E30. The definition of this notation is presented in "Statecharts: A Visual Formalism for Complex Systems"[1].

7.2.2 Transition tables are provided in conjunction with the state diagrams to explicitly describe the nature of each state transition. A transition table contains columns for Transition number, Previous State, Trigger, New State, Actions, and Comments. The 'trigger' (column 3) for the transition occurs while in the 'previous' state. The 'actions' (column 5) includes a combination of:

1. Actions taken upon exit of the previous state.
2. Actions taken upon entry of the new state.
3. Actions taken that are most closely associated with the transition.

7.2.3 No differentiation is made between these cases.

#	*Previous State*	*Trigger*	*New State*	*Actions*	*Comments*
Transition #					

7.3 *Variable Data Item Definition*

7.3.1 The variable item dictionary contained in this Standard defines the variable name, description, type (format), access (read/write or read only), and comments about the variable. This is depicted in the format of the following table:

Variable Name	*Description*	*Type*	*Access*	*Comment*

7.4 *Event Description*

7.4.1 The collection event description table contained in this Standard includes the event name, the state model transition, the description, and the data required for the event. This is depicted in the format of the following table:

Event Name	*State Model Transition Number*	*Description*	*Required Data*

8 Overview

8.1 This section defines the concepts that enable EPT to provide information required for determining equipment performance.

8.2 Equipment processing times, including throughput and cycle times, are determined by two factors: (1) the sequence of tasks performed by the equipment's EPT modules, including the number of each task performed, as well as the series and parallel relationships between tasks, and (2) the execution time of each individual task. By tracking equipment systems in terms of modules that perform tasks, it will be possible to precisely identify the effect of both factors toward overall system performance.

[1] Harel, D., "Statecharts: A Visual Formalism for Complex Systems," Science of Computer Programming 8 (1987): pp. 231–274.

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8.3 EPT enables tracking to this level of detail by introducing the concept of an EPT module and the concept of a task. EPT-compliant equipment are required to report events according to a breakdown of equipment operations into tasks and EPT modules. An EPT module is an equipment component that affects processing or throughput, and is capable of executing only one task at a time. Examples of EPT modules include process stations, process chambers, wafer handling systems, etc. A task is a planned and repeatable activity with an expected duration and a definite beginning and end. Examples of tasks include the following:

- Move wafer from cassette to stage,
- Pre-align wafer,
- Align reticle,
- Preheat chamber,
- Increase vacuum etc.

8.4 EPT-compliant equipment consist of EPT modules that perform tasks. Thus, the state of the equipment can be effectively modeled as a composition of the states of the EPT modules. In addition, EPT provides detailed information required for performance tracking by providing the task-level details for each EPT module.

8.5 EPT modules and tasks are to be defined such that each EPT module executes only one task at a time. An example is given in Related Information 1, § R1-8. Each equipment system must have defined at least as many EPT modules as tasks that may be performed simultaneously by that system.

8.6 Each task should be defined such that the units involved (e.g., wafers, substrates, reticles), the EPT modules involved (e.g., process station, robot, etc.), and the task type (e.g., ‘process’, ‘support’, etc.) are constant throughout the duration of the task. A change in any of the following items should result in the start of a new task or completion of current task:

- Change in material involved (e.g., arrival of wafer, removal of wafer)
- Change of EPT modules involved (e.g., robot moves wafer, chamber processes wafer)
- Change in task type (e.g., purge recipe followed by production recipe)

NOTE 1: It is not practical for equipment to define tasks or modules at an indefinite level of detail. It is therefore recommended that in each instance for which the material involved, the EPT modules involved, and the task types are all constant, events for only one task should be reported. Any further breakdown is not required for EPT compliance. EPT places no limitations on supplemental event reporting within any task. However, these supplemental events should not be reported as EPT events.

9 EPT State Model

9.1 This section defines the basic Equipment Performance Tracking State Model, which is applied to the equipment and its EPT modules.

9.2 *EPT State Model Requirements*

9.2.1 The EPT state model is intended to capture the different states of the equipment and its EPT modules from an operational point of view:

- EPT module busy executing a task.
- EPT module blocked from executing a task.
- EPT module idle.

9.2.2 The equipment shall maintain an EPT state model for each major module (e.g., process stations and chambers, wafer handling systems, etc.) through which a unit passes throughout the entire equipment from removal of units from the carrier to the return of units to the carrier.

9.2.3 The equipment shall maintain an EPT state model for any major component of the equipment that impacts processing and throughput.

9.2.4 The equipment shall maintain its own EPT state model for the overall equipment.

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9.2.5 The equipment is responsible for communicating all EPT state transitions.

9.2.6 The EPT state model can be applied to the Equipment Front-End Module (EFEM).

9.2.7 There are two types of EPT Modules: (1) Production EPT modules, and (2) EFEM/LoadPort EPT modules. Both types have a module-level EPT State Machine.

9.3 *EPT State Model Diagram*

9.3.1 Figure 1 shows the EPT state module diagram.

9.4 *EPT State Definitions*

9.4.1 This section describes the EPT states.

9.4.2 *IDLE*

9.4.2.1 *EPT Module* — The following conditions are required for the module to be in the IDLE state:

- No material is present in the module,
- The module is not executing a task, and
- No fault condition exists in the module that prevents it from starting a new task.

9.4.2.2 *Equipment* — The following conditions are required for the equipment to be in the IDLE state:

- No BLOCKED EPT modules prevent an EPT module from starting a new task, and
- There is no EPT module in the BUSY state with a task type of 1, 2, 3, 4, or 5.

9.4.2.3 For both the equipment and modules the following conditions may exist in the IDLE State:

- The equipment or module is maintaining and monitoring environmental requirements (e.g., background temperature, particle monitoring, etc.).

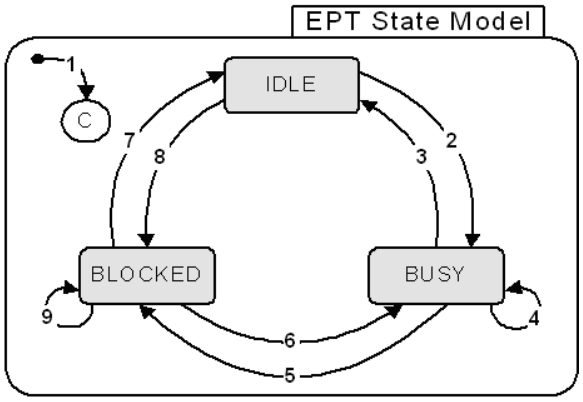

Figure 1
Equipment Performance Tracking State Model

9.4.3 *BUSY*

9.4.3.1 *EPT Module* — The following conditions are required for the module to be in the BUSY state:

- The module is executing a task, and
- No fault condition exists in the module that prevents the execution of a task.

9.4.3.2 *Equipment* — The following condition is required for the equipment to be in the BUSY state:

- At least one EPT module is BUSY with a task type of 1, 2, 3, 4, or 5.

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9.4.3.3 For both the equipment and modules the following conditions may exist in the BUSY State:

- Material is present.

9.4.4 *BLOCKED*

9.4.4.1 *EPT Module* — One or more of the following conditions are required for the module to be in the BLOCKED state:

- Conditions exist that do not allow the EPT module to continue or start execution of a task.
- A fault condition(s) exists that prevents the EPT module from completing its task.
- A fault condition(s) exists that prevents the EPT module from starting a new task.
- The EPT module is pausing (or aborting) as the result of a pause (or abort) directive.
- The EPT module has paused its task and is awaiting a resume directive.
- The EPT module fails to initialize upon start-up.

9.4.4.2 *Equipment* — The following condition is required for the EPT equipment to be in the BLOCKED state:

- No EPT module is BUSY with task type of 1, 2, 3, 4, or 5, and
- At least one BLOCKED EPT module prevents any EPT module from starting a new task or completing current task.

9.4.4.3 For both the EPT equipment and modules the following conditions may exist in the BLOCKED State:

- Material is present.

9.5 *EPT State Transition Table — Module Level*

9.5.1 This section defines the state transitions for the EPT modules within the equipment.

9.5.2 For each status variable updated with the event change, the corresponding event variable shall also be updated (refer to Table 4).

Table 1 Basic EPT State Transitions – Module Level

#	*PREVIOUS STATE*	*TRIGGER*	*NEW STATE*	*ACTIONS*	*COMMENTS*
1	(No state)	EPT Module$_i$ initialization completed.	IDLE [OR] BLOCKED [OR] BUSY	No EPT event is triggered • *EPTState* = IDLE or BLOCKED or BUSY • *PreviousEPTState* = NoState • *PreviousTaskName* = 'No Task' • *PreviousTaskType* = 0 • *EPTStateTime* = 0 • *BlockedReason* may be updated • *BlockedReasonText* may be updated • *TaskName* is set • *TaskType* is set	Note that (no state) is not an actual EPT state. It is shown here simply to indicate that a new EPT state is entered upon completion of initialization. If no Fault conditions exist and no material is present, transition to IDLE or BUSY. If Fault conditions exist, or EPT module cannot be initialized, then transition to BLOCKED. If Material is present, then transition to BLOCKED or BUSY.

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#	*PREVIOUS STATE*	*TRIGGER*	*NEW STATE*	*ACTIONS*	*COMMENTS*
2	IDLE	EPT Module$_i$ starts execution of a new task.	BUSY	• *EPTStateChange$_i$* event is triggered • *EPTStateTime* is calculated • *EPTState* = BUSY • *PreviousEPTState* = IDLE • *TaskName* is set • *TaskType* is set	
3	BUSY	EPT Module$_i$ completes execution of task. [AND] Material is removed from the EPT Module$_i$.	IDLE	• *EPTStateChange$_i$* event is triggered • *EPTStateTime* is calculated • *EPTState* = IDLE • *PreviousEPTState* = BUSY • *TaskName* = 'No Task' • *TaskType* = 0 • *PreviousTaskName* is set • *PreviousTaskType* is set	
4	BUSY	EPT Module$_i$ starts execution of a new task upon the normal completion of the previous task.	BUSY	• *EPTStateChange$_i$* event is triggered • *TaskName* is set • *TaskType* is set • *PreviousTaskName* is set • *PreviousTaskType* is set	EPTStateTime does not change. Thus, when the module transitions out of BUSY state, the value of EPTStateTime will reflect the total time the module was BUSY. PreviousEPTState does not change. Thus, PreviousEPTState will continue to reflect the state of the module before it was BUSY.
5	BUSY	A pause command is received that pauses the EPT Module$_i$. [OR] An abort command is received that aborts the EPT Module$_i$. [OR] A Fault condition occurs that prevents the EPT Module$_i$ from executing its task.	BLOCKED	• *EPTStateChange$_i$* event is triggered • *EPTStateTime* is calculated • *BlockedReason* is set • *BlockedReasonText* is set • *EPTState* = BLOCKED • *PreviousEPTState* = BUSY • *PreviousTaskName* is set • *PreviousTaskType* is set	TaskName and TaskType do not change. The variables continue to reflect the information of the task that is currently blocked from executing.

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#	*PREVIOUS STATE*	*TRIGGER*	*NEW STATE*	*ACTIONS*	*COMMENTS*
6	BLOCKED	All fault conditions that prevented task execution are cleared and the EPT Module$_i$ resumes execution of its task or starts execution of a new task. [OR] A resume command is received that enables the EPT Module$_i$ to resume its task.	BUSY	• *EPTStateChange$_i$* event is triggered • *EPTStateTime* is calculated • *EPTState* = BUSY • *PreviousEPTState* = BLOCKED • *TaskName* is set • *TaskType* is set • *BlockedReason* = 0 • *BlockedReasonText* = 'Not Blocked'	
7	BLOCKED	All fault conditions that prevented task execution are cleared. [AND] All material is removed from the EPT Module$_i$. [AND] EPT Module$_i$ can begin a new task.	IDLE	• *EPTStateChange$_i$* event is triggered • *EPTStateTime* is calculated • *EPTState* = IDLE • *PreviousEPTState* = BLOCKED • *TaskName* = 'No Task' • *TaskType* = 0 • *PreviousTaskName* is set • *PrevoiusTaskType* is set • *BlockedReason* = 0 • *BlockedReasonText* = 'Not Blocked'	
8	IDLE	Material arrives at the EPT Module$_i$ and the EPT Module$_i$ cannot begin executing a task on the material. [OR] A fault condition occurs which prevents the EPT Module$_i$ from starting a new task.	BLOCKED	• *EPTStateChange$_i$* event is triggered • *EPTStateTime* is calculated • *BlockedReason* is set • *BlockedReasonText* is set • *EPTState* = BLOCKED • *PreviousEPTState* = IDLE	
9	BLOCKED	Fault conditions occur that prevent the EPT Module$_i$ from resuming a blocked task, or starting a new task.	BLOCKED	• *EPTStateChange$_i$* event is triggered • *BlockedReason* is set • *BlockedReasonText* is set	EPTStateTime does not change. Thus, when the module transitions out of BLOCKED state, the value of EPTStateTime will reflect the total time the module was BLOCKED from processing. PreviousEPTState does not change. Thus, PreviousEPTState will continue to reflect the state of the module before it was BLOCKED from processing.

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9.6 *EPT State Transition Table for Equipment*

9.6.1 This section defines the State Transitions for the Equipment, in terms of its EPT modules.

Table 2 Basic EPT State Transitions — Equipment Level

#	*Previous State*	*Trigger*	*New State*	*Actions*	*Comments*
1	(No state)	Equipment initiation completed (includes initialization of EPT modules).	IDLE [OR] BLOCKED [OR] BUSY	• *EPTStateChange$_0$* event is triggered • *EPTState* = IDLE or BLOCKED or BUSY • *PreviousEPTState* = NoState • *EPTStateTime* = 0 • *BlockedReason* may be updated • *BlockedReasonText* may be updated	
2	IDLE	At least one EPT module transitions to the BUSY state with task type of 1..5.	BUSY	• *EPTStateChange$_0$* event is triggered • *EPTStateTime* is calculated • *EPTState* = BUSY • *PreviousEPTState* = IDLE	
3	BUSY	All EPT modules that were BUSY with task type of 1..5 transition either to IDLE or to BUSY with task type other than 1..5.	IDLE	• *EPTStateChange$_0$* event is triggered • *EPTStateTime* is calculated • *EPTState* = IDLE • *PreviousEPTState* = BUSY	
4	BUSY	Any EPT module transitions to BUSY state with task type 1..5.	BUSY		

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#	*Previous State*	*Trigger*	*New State*	*Actions*	*Comments*

5	BUSY	{
		At least one EPT module transitions to a BLOCKED state and prevents another EPT module from starting a new task or completing an existing task.
		
		[AND]
		
		There are no other EPT modules BUSY with task type 1..5.
		
		} **[OR]** {
		
		Any EPT module that was BUSY with task type 1..5 transitions to either IDLE or BUSY with task type other than 1..5.
		
		[AND]
		
		There is at least one EPT module in BLOCKED state that is preventing another EPT module from starting a new task or completing an existing task.
		}
			BLOCKED	• *EPTStateChange*$_0$ event is triggered
				• *EPTStateTime* is calculated
				• *EPTState* = BLOCKED
				• *PreviousEPTState* = BUSY
				• *BlockedReason* is set
				• *BlockedReasonText* is set	

6	BLOCKED	At least one EPT module transitions to BUSY with task type of 1..5.	BUSY	• *EPTStateChange*$_0$ event is triggered • *EPTStateTime* is calculated • *EPTState* = BUSY • *PreviousEPTState* = BLOCKED	
7	BLOCKED	All BLOCKED EPT modules that prevented another EPT module from starting a new task or completing a current task no longer prevents the EPT module from doing so. **[AND]** No EPT module is BUSY with task type 1..5.	IDLE	• *EPTStateChange*$_0$ event is triggered • *EPTStateTime* is calculated • *EPTState* = IDLE • *PreviousEPTState* = BLOCKED	
8	IDLE	Any EPT module transitions to BLOCKED that prevents another EPT module from starting a new task.	BLOCKED	• *EPTStateChange*$_0$ event is triggered • *EPTStateTime* is calculated • *BlockedReason* is set • *BlockedReasonText* is set • *EPTState* = BLOCKED • *PreviousEPTState* = IDLE	

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#	*Previous State*	*Trigger*	*New State*	*Actions*	*Comments*
9	BLOCKED	Any EPT module transitions to BLOCKED that prevents or continues to prevent another EPT module from starting a new task or completing a current task.	BLOCKED	• *EPTStateChange*$_0$ event is triggered • *EPTStateTime* is calculated • *BlockedReason* is set • *BlockedReasonText* is set	EPTStateTime does not change. Thus, when the equipment transitions out of BLOCKED state, the value of EPTStateTime will reflect the total time the equipment was BLOCKED from processing. PreviousEPTState does not change. Thus, PreviousEPTState will continue to reflect the state of the equipment before it was BLOCKED from processing.

10 EPTTracker Objects

10.1 The equipment shall provide SEMI E39 compliant EPTTracker objects to track performance of EPT Equipment and Modules. The equipment shall provide one EPTTracker object for the equipment and one EPTTracker object for each EPT Module (both Production and EFEM/LoadPort EPT Modules). The EPTTracker object is used to maintain and update the SEMI E116 attributes each time a SEMI E116 transition occurs. These EPTTracker objects will be owned at the equipment level. These objects shall be available immediately after equipment initialization, and shall be persistent until equipment shutdown. If certain attributes are identified as persistent, the object shall restore the value of these attributes on initialization. The Host can use SEMI E39 services (e.g., GetAttr) to access EPTTracker objects. The EPTTracker object format is specified in Table 3.

Table 3 EPTTracker Object Definition

Attribute Name	*Definition*	*Access*	*Reqd*	*Form*
ObjType	Object type	RO	Y	Text ‘EPTTracker’
ObjID	Object identifier. A name assigned by the Equipment designer uniquely identifying the instance of the EPTTracker object within this Equipment.	RO	Y	Text 1–80 characters. Shall conform to SEMI E39 restrictions on ObjID.
BlockedReason	A numeric code that identifies the most recent blocked condition of the related EPT module.	RO	Y	Enumerated: 0 = Not Blocked 1 = Unknown 2 = Safety Threshold 3 = Error Condition 4 = Parametric Exception 5 = Aborting, Aborted 6 = Pausing, Paused 7 = Reserved 8 = Reserved 9 = Reserved

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Attribute Name	*Definition*	*Access*	*Reqd*	*Form*
BlockedReasonText	A description of the most recent blocked condition of this Equipment or EPT module. May provide further details to BlockedReason.	RO	Y	Text 0–80 characters This attribute must be defined for both equipment and module EPTTracker objects. For the Equipment, a BlockedReason value is determined by the blocked module.
DisableEventOnTransition	A list of zero to nine EPT State Model transitions for which event reporting to the host are disabled.	RW	Y	List of zero to nine integers in the range 1–9.
EPTElementType	A numeric code that indicates whether this EPTTracker object refers to the equipment or an EPT module.	RO	Y	Enumerated 0 = Equipment 1 = Production EPT Module 2 = EFEM/LoadPort EPT Module
EPTState	The current EPT state of this equipment or EPT module.	RO	Y	Enumerated 0 = Idle 1 = Busy 2 = Blocked
EPTStateTime	Time spent in the previous EPT State, prior to entering the current EPTState.	RO	Y	Unsigned integer Value in seconds
EPTElementName	A human-understandable name for this equipment or module, which must be unique among all objects of EPTTracker type.	RO	Y	Text 1–80 characters
PreviousEPTState	The previous EPT state of this equipment or EPT module, prior to entering the current EPTState.	RO	Y	Enumerated 0 = Idle 1 = Busy 2 = Blocked 3 = No State
PreviousTaskName	Name of the EPT Task previously running on this EPT module, prior to starting the current EPT Task.	RO	Y	Text 0–80 characters This attribute must be defined for both equipment and module EPTTracker objects. For the Equipment, the value of the PreviousTaskName attribute may be left blank (empty string).

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Attribute Name	*Definition*	*Access*	*Reqd*	*Form*
PreviousTaskType	Type of the EPT Task previously running on this EPT module, prior to starting the current EPT Task.	RO	Y	Enumerated: 0 = No Task 1 = Unspecified 2 = Process – adding value (e.g., exposing) 3 = Support – incapable of adding value (e.g., Handling/Transport) 4 = Equipment Maintenance (e.g., Equipment-initiated clean cycle) 5 = Equipment Diagnostics (e.g., Equipment-initiated health check) 6 = Waiting (e.g., chamber waiting for a robot to remove a substrate) This attribute must be defined for both equipment and module EPTTracker objects. For the Equipment, the value of the PreviousTaskType attribute may be set to 0.
TaskName	Name of the EPT Task currently running on this EPT module.	RO	Y	Text 0–80 characters This attribute must be defined for both equipment and module EPTTracker objects. For the Equipment, the value of the TaskName attribute may be left blank (empty string).
TaskType	The type of EPT Task currently running on this EPT module.	RO	Y	Enumerated: 0 = No Task 1 = Unspecified 2 = Process – adding value (e.g., exposing) 3 = Support – incapable of adding value (e.g., Handling/Transport) 4 = Equipment Maintenance (e.g., Equipment-initiated clean cycle) 5 = Equipment Diagnostics (e.g., Equipment-initiated health check) 6 = Waiting (e.g., chamber waiting for a robot to remove a substrate) This attribute must be defined for both equipment and module EPTTracker objects. For the Equipment, the value of the TaskType attribute may be set to 0.
TransitionTimestamp	The timestamp when the most recent transition occurred and collection event was triggered that set the tracker to its current state.	RO	Y	Formatted text: Timestamp format as specified by the TimeFormat equipment constant value setting (refer to SEMI E5).
Transition	The most recent transition that updated the tracker.	RO	Y	Enumerated: 1 = Transition 1 2 = Transition 2 … 9 = Transition 9
TrackerEventID	Numeric identifier for the event that is triggered for a change in the state model for this tracker.	RO	Y	Unsigned Integer

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10.2 The object attributes *TaskType* and *PreviousTaskType* may take on one of the following values:

0. *No Task* — Value for *TaskType* when the EPT module is IDLE, and value for the *PreviousTaskType* when the EPT module transitions out of the IDLE state or initial state (no state).
1. *Unspecified* — A placeholder for unspecified task types.
2. *Process* — Tasks expected under normal or desired manufacturing operations that are capable of adding value by either (1) a physical-chemical change or (2) providing critical process information (e.g., inspection, metrology). This includes tasks that are dependent on a recipe ID, dependent on parameters (e.g., target etch thickness), or recipe and parameter independent (e.g., fixed purge time for all recipes or regardless target etch thickness).
3. *Support* — Tasks expected under normal or desired manufacturing operations that are incapable of adding value according to the criteria for a process task. These include alignment, handling and other transport, and environmental changes by supporting EPT modules (e.g., pump down and vent operations in a load lock).
4. *Maintenance* — Any task that is intended to change the state of the equipment for the purpose of maintaining equipment functionality or performance. This includes clean cycles, purges not expected under normal or desired manufacturing operations, and certain reset operations that do not put the EPT module or equipment in a non-operational state.
5. *Diagnostics* — Any task that obtains information about the status of the equipment for the purpose of determining the equipment's health or identifying an equipment problem(s). This excludes any task that is intended to change the status of the equipment, as in a maintenance task. This also excludes metrology and inspection tasks that provide critical information about the process even if those tasks also provide equipment diagnostic information as a byproduct.
6. *Waiting* — An equipment-initiated task that indicates the module is encountering normal waiting periods as part of its normal wafer processing scenario. A 'Waiting' task applies when the module is awaiting a task to be completed by an external source before it can continue/complete processing (e.g., processing chamber waiting for robot to remove wafer, loadport waiting for host ID/slot map verification, module waiting for a host command to initiate processing and commence material unloading). Material must be present in the module when this task occurs.

10.2.1 *NOTES* — The following notes resolve ambiguities that arise from limitations of the equipment or EPT module point of view. From the equipment or EPT module point of view:

- Process maintenance activities (e.g., monitor wafers) and engineering activities (e.g., new process qualification runs) may be indistinguishable from normal or desired manufacturing operations. These operations shall be reported as either process or support operations.
- Equipment maintenance activities (e.g., equipment qualification runs monitor wafers) may be indistinguishable from normal or desired manufacturing operations. These operations shall be reported as either process or support operations.
- Activities executed for exception handling (e.g., pausing, aborting, stopping, or otherwise moving to a safe state) may be indistinguishable from support tasks. Examples include handling/transporting wafers to safe locations, pumping, venting, purging, etc. If the equipment or module implementing EPT does not have *reasonable inherent knowledge of purpose* regarding these activities, the tasks should be always declared as support tasks. If the equipment or module implementing EPT does have reasonable inherent knowledge of purpose regarding these activities, these activities shall be either (1) modeled as task faults or otherwise blocked conditions, or (2) exempted entirely from treatment under EPT. In case 2, the affected EPT module or modules must be in a BLOCKED state due to related conditions reported to the EPT state model(s).

10.3 *Blocked Reason Text Requirements*

- Fault Condition: Blocked reason text should begin with "Fault: - 'appropriate text' "
- Pause Condition: Blocked reason text should begin with "Pause: - 'appropriate text' "
- Abort Condition: Blocked reason text should begin with "Abort: - 'appropriate text' "

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10.4 *Configurable Disabling of Event Reporting for Selected Transitions*

10.4.1 The attribute DisableEventOnTransition may be used by the host to selectively disable EPT state model transition events for an individual EPTTracker object independently of whether the events are enabled at the equipment or not. This attribute is a simple list of integers 1–9 representing the disabled events.

10.4.2 For example, a list {1,4} indicates that no event reports should be sent for transitions 1 and 4. This attribute may be set by using the SEMI E39 service SetAttr for one or more objects.

10.4.3 The DisableEventOnTransition attribute also applies to the transition-specific events.

11 EPT Events

11.1 This section defines the required events and data variables associated with transitions in the EPT state model.

11.2 Table 4 defines the data variables that are valid at the time of an EPT State Change event. When the equipment or EPT module triggers an EPTStateChange event, the value of the data variable shall be updated to reflect the data valid at the time the event occurred.

Table 4 Data Variable Definitions

Variable Name	*Description*	*Type*	*Access*	*Comment*
EPTState	The new (resulting) EPT state of the module or equipment at the end of an EPT state transition.	Enumerated	RO	0 = IDLE 1 = BUSY 2 = BLOCKED
PreviousEPTState	The previous EPT state of the module or equipment of an end of an EPT state transition.	Enumerated	RO	0 = IDLE 1 = BUSY 2 = BLOCKED 3 = No State
EPTStateTime	Time spent in previous EPT state for the equipment or module.	Unsigned integer	RO	Seconds This is the time period between the entry into the previous EPT state and the entry into the current EPT State.
TaskName	Name of the task that was started by the module or equipment at the occurrence of an EPT state transition.	Text, 0–80 characters	RO	TaskName is defined by the equipment supplier.
TaskType	The type of EPT Task currently running on this EPT module.	Enumerated	RO	0 = No Task 1 = Unspecified 2 = Process – adding value (e.g., exposing) 3 = Support – incapable of adding value (e.g., Handling/Transport) 4 = Equipment Maintenance (e.g., equipment initiated clean cycle) 5 = Equipment Diagnostics (e.g., equipment-initiated health check) 6 = Waiting (e.g., chamber waiting for a robot to remove a substrate)
PreviousTaskName	Name of the task that was completed by the module or equipment at the occurrence of an EPT state transition.	Text, 0–80 characters	RO	

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Variable Name	*Description*	*Type*	*Access*	*Comment*
PreviousTaskType	The type of the task that was completed by the module or equipment at the occurrence of an EPT state transition.	Enumerated	RO	0 = No Task 1 = Unspecified 2 = Process – adding value (e.g., exposing) 3 = Support – incapable of adding value (e.g., Handling/Transport) 4 = Equipment Maintenance (e.g., equipment initiated clean cycle) 5 = Equipment Diagnostics (e.g., equipment-initiated health check) 6 = Waiting (e.g., chamber waiting for a robot to remove a substrate)
BlockedReason	The number of the block condition that initiated the transition to the BLOCKED state for the module or equipment.	Enumerated	RO	0 = Not Blocked 1 = Unknown 2 = Safety Threshold 3 = Error Condition 4 = Parametric Exception 5 = Aborting, Aborted 6 = Pausing, Paused 7 = Reserved 8 = Reserved 9 = Reserved
BlockedReasonText	The reason why the transition was made to the BLOCKED state for the module or equipment.	Text, 0–80 characters	RO	Description of the Blocked reason
EPTObjID	Object identifier. A name assigned by the Equipment designer uniquely identifying the instance of the EPTTracker object within this Equipment.	Text, 1-80 characters	RO	Corresponds to the EPT Tracker object's ObjID attribute.
EPTElementName	A human-understandable name for this equipment or module, which must be unique among all objects of EPTTracker type.	Text, 1-80 characters	RO	Corresponds to the EPT Tracker object's EPTElementName attribute.
EPTElementType	A numeric code that indicates whether this EPTTracker object refers to the equipment or an EPT module.	Enumerated	RO	Corresponds to the EPT Tracker object's EPTElementType attribute. Uses the same enumeration values as attribute EPTElementType.

11.3 All collection events require the following data variables to be available:

<Clock> (*Clock* represents the timestamp for the occurrence of the event)

<EPTState>

<PreviousEPTState>

<EPTStateTime>

<TaskName>

<TaskType>

<PreviousTaskName>

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<PreviousTaskType>

<EPTObjID>

<EPTElementName>

<EPTElementType>

Additional data variables required for specific transitions are shown in Table 5.

Table 5 Events

Event Name	*State Model Transition Number*	*Description*	*Additional Required Data Variables*
EPTStateChange$_0$	1, 2, 3, 4, 5, 6, 7, 8, 9	Triggered by an EPT State change at the equipment-level.	The following additional data is required for Transitions 5, 8, and 9: <BlockedReason> <BlockedReasonText>
EPTStateChange$_i$	1, 2, 3, 4, 5, 6, 7, 8, 9	Triggered by an EPT State change at the EPT module-level. There exist i+1 EPTStateChange events, where i is the number of EPT modules. EPTStateChange$_1$ is triggered by a state change at the 1st EPT module, EPTStateChange$_2$ is triggered by a state change at the 2nd EPT module, etc.	The following additional data is required for Transitions 5, 8, and 9: <BlockedReason> <BlockedReasonText>

11.4 *EPT Transition-Specific Events*

11.4.1 The EPT Transition-Specific Events allow a host to be notified only when specific transitions occur and to setup different event reports for each transition.

11.4.2 The EPT Transition-Specific Events can be implement concurrently with the EPTStateChange events in Table 5.

11.4.3 The same set of EPTModuleTransition collection events is used for all modules. The host can distinguish the collection events corresponding to each module using the EPTObjID or EPTElementName data variable in an event report.

Table 6 Transition-Specific Events

Event Name	*State Model Transition Number*	*Description*	*Additional Required Data Variables*
EPTEquipmentTransition1	1	Triggered by an EPT transition 1 State change at the equipment-level.	
EPTEquipmentTransition2	2	Triggered by an EPT transition 2 State change at the equipment-level.	
EPTEquipmentTransition3	3	Triggered by an EPT transition 3 State change at the equipment-level.	
EPTEquipmentTransition4	4	Triggered by an EPT transition 4 State change at the equipment-level.	
EPTEquipmentTransition5	5	Triggered by an EPT transition 5 State change at the equipment-level.	<BlockedReason> <BlockedReasonText>
EPTEquipmentTransition6	6	Triggered by an EPT transition 6 State change at the equipment-level.	

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Event Name	*State Model Transition Number*	*Description*	*Additional Required Data Variables*
EPTEquipmentTransition7	7	Triggered by an EPT transition 7 State change at the equipment-level.	
EPTEquipmentTransition8	8	Triggered by an EPT transition 8 State change at the equipment-level.	<BlockedReason> <BlockedReasonText>
EPTEquipmentTransition9	9	Triggered by an EPT transition 9 State change at the equipment-level.	<BlockedReason> <BlockedReasonText>
EPTModuleTransition1	1	Triggered by an EPT transition 1 State change at the EPT module-level.	
EPTModuleTransition2	2	Triggered by an EPT transition 2 State change at the EPT module-level.	
EPTModuleTransition3	3	Triggered by an EPT transition 3 State change at the EPT module-level.	
EPTModuleTransition4	4	Triggered by an EPT transition 4 State change at the EPT module-level.	
EPTModuleTransition5	5	Triggered by an EPT transition 5 State change at the EPT module-level.	<BlockedReason> <BlockedReasonText>
EPTModuleTransition6	6	Triggered by an EPT transition 6 State change at the EPT module-level.	
EPTModuleTransition7	7	Triggered by an EPT transition 7 State change at the EPT module-level.	
EPTModuleTransition8	8	Triggered by an EPT transition 8 State change at the EPT module-level.	<BlockedReason> <BlockedReasonText>
EPTModuleTransition9	9	Triggered by an EPT transition 9 State change at the EPT module-level.	<BlockedReason> <BlockedReasonText>

12 EPT Status Variables

12.1 The following subscripted status variables shall be defined for each EPT Module, where ‘i’ corresponds to the ith module starting with ‘1’.

12.2 Each subscripted status variable with the same index ‘i’ corresponds to the same EPT Module.

12.3 A set of subscribed status variables with the index ‘0’ shall be defined to reflect the Equipment EPT module.

12.4 The value for each status variable shall be equivalent to the value of the EPT Tracker object’s corresponding attribute value.

Table 7 Status Variable Definitions Representing Object Attributes

Name	*Description*	*Format*	*Comment*
BlockedReason_i	Corresponds to the EPT Tracker object’s BlockedReason attribute.	Enumerated	Same enumeration as BlockedReason attribute
BlockedReasonText_i	Corresponds to the EPT Tracker object’s BlockedReasonText attribute.	Text, 0–80 characters	
Clock	Timestamp for the state transition.	Formatted numeric text: timestamp format	When included in an event report, Clock represents the timestamp for the occurrence of the event and transition.

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Name	*Description*	*Format*	*Comment*
EPTElementName_i	Corresponds to the EPT Tracker object's EPTElementName attribute.	Text, 1-80 characters	
EPTElementType_i	Corresponds to the EPT Tracker object's EPTElementType attribute.	Enumerated	Same enumeration as EPTElementType attribute
EPTObjID_i	Corresponds to the EPT Tracker object's ObjID attribute.	Text, 1-80 characters	
EPTState_i	Corresponds to the EPT Tracker object's EPTState attribute.	Enumerated	Same enumeration as EPTState attribute
EPTStateTime_i	Corresponds to the EPT Tracker object's EPTStateTime attribute.	Unsigned integer	Seconds
NumberOfEPTModules	The number of EPT modules in the equipment. This does not include the equipment EPT.	Unsigned Integer	
PreviousEPTState_i	Corresponds to the EPT Tracker object's PreviousEPTState attribute.	Enumerated	Same enumeration as PreviousEPTState attribute
PreviousTaskName_i	Corresponds to the EPT Tracker object's PreviousTaskName attribute.	Text, 0–80 characters	
PreviousTaskType_i	Corresponds to the EPT Tracker object's PreviousTaskType attribute.	Enumerated	Same enumeration as PreviousTaskType attribute
TaskName_i	Corresponds to the EPT Tracker object's TaskName attribute.	Text, 0–80 characters	
TaskType_i	Corresponds to the EPT Tracker object's TaskType attribute.	Enumerated	Same enumeration as TaskType attribute
TrackerEventID_ i	Corresponds to the EPT Tracker object's TrackerEventID attribute.	Unsigned Integer	
Transition_ i	Corresponds to the EPT Tracker object's Transition attribute.	Enumerated	Same enumeration as Transition attribute
TransitionTimestamp_ i	Corresponds to the EPT Tracker object's TransitionTimestamp attribute.	Text, format as specified by the TimeFormat equipment constant value setting (refer to SEMI E5).	

13 Well-Known Names

13.1 *Background*

13.1.1 Implementers of this Specification may choose any names for variables, collection events and alarms. For example, SEMI E116 defines a collection event for all Equipment Performance Tracking transitions. Implementers might define the CENAME for this collection event as EPTStateChange, E116_EPTStateChange, EPTTransition, or CollectionEvent116. All of these CENAME are valid.

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13.1.2 For end users, it can be useful to map the names for variables, collection events and alarms in a specific implementation to the variables, collection events and alarms defined in this Specification. Well-known names are standardized names for the variables, collection events and alarms defined in this Specification.

13.1.3 Equipment implementing this Specification can publish a map between the names in their implementation and the well-known names defined in this section using a SEMI E172 SECS Equipment Data Dictionary (SEDD) file. The map can also be published in the SEMI E30 documentation.

13.1.4 New implementations can use well-known names such as the SECS item name (CENAME, SVNAME, etc.).

13.1.5 The list of well-known variables, collection events and alarms in this section does not intend to imply that the implementations of this Specification must include all the listed variables, collection events and alarms.

13.1.6 For completeness, well-known tables for variables, collection events and alarms included yet deliberately left empty when no applicable variable, collection event, or alarm is defined in this Specification.

13.1.7 A '<#>' in the well-known name means that there can be multiple instances. Sufficient leading zeros should be added to allow logical alphabetical sorting. For example, E116:EPTState.01, E116:EPTState.02, …, E116:EPTState.88, if there are up to 99 instances.

13.1.8 For the '_i' well known, the '_0' instance is distinguished by a separate well-known name without a '<#>' suffix. It represents the unique EPT Equipment instance. The other '_i' instances greater than zero represent the EPT Modules where a '.<#>' suffix is specified in the well-known name.

13.1.9 Well-known names are case-sensitive.

13.2 *Well-Known Names for Variables*

13.2.1 This section defines well-known names for variables currently defined in this Specification.

13.2.2 Refer to SEMI E5 for more information on the variable classes.

13.2.3 When a variable corresponds to an object's attribute, the well-known name uses the naming convention 'E116:<object name>:<attribute name>' where '<object name>' is the name of the object without spaces, '<attribute name>' is name of the corresponding attribute without spaces.

Table 8 Well-Known Names for Variables

Variable	*Variable Class*	*Well-Known Name*
EPTState	DV	E116:EPTTracker:EPTState
PreviousEPTState	DV	E116:EPTTracker:PreviousEPTState
EPTStateTime	DV	E116:EPTTracker:EPTStateTime
TaskName	DV	E116:EPTTracker:TaskName
TaskType	DV	E116:EPTTracker:TaskType
PreviousTaskName	DV	E116:EPTTracker:PreviousTaskName
PreviousTaskType	DV	E116:EPTTracker:PreviousTaskType
BlockedReason	DV	E116:EPTTracker:BlockedReason
BlockedReasonText	DV	E116:EPTTracker:BlockedReasonText
EPTObjID	DV	E116:EPTTracker:ObjID
EPTElementName	DV	E116:EPTTracker:EPTElementName
EPTElementType	DV	E116:EPTTracker:EPTElementType
BlockedReason_0	SV	E116:EPTTracker:BlockedReason
BlockedReason_i	SV	E116:EPTTracker:BlockedReason.<#>
BlockedReasonText_0	SV	E116:EPTTracker:BlockedReasonText
BlockedReasonText_i	SV	E116:EPTTracker:BlockedReasonText.<#>
EPTElementName_0	SV	E116:EPTTracker:EPTElementName

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Variable	*Variable Class*	*Well-Known Name*
EPTElementName_i	SV	E116:EPTTracker:EPTElementName.<#>
EPTElementType_0	SV	E116:EPTTracker:EPTElementType
EPTElementType_i	SV	E116:EPTTracker:EPTElementType.<#>
EPTObjID_0	SV	E116:EPTTracker:ObjID
EPTObjID_i	SV	E116:EPTTracker:ObjID.<#>
EPTState_0	SV	E116:EPTTracker:EPTState
EPTState_i	SV	E116:EPTTracker:EPTState.<#>
EPTStateTime_0	SV	E116:EPTTracker:EPTStateTime
EPTStateTime_i	SV	E116:EPTTracker:EPTStateTime.<#>
NumberOfEPTModules	SV	E116:EPTTracker:NumberOfEPTModules
PreviousEPTState_0	SV	E116:EPTTracker:PreviousEPTState
PreviousEPTState_i	SV	E116:EPTTracker:PreviousEPTState.<#>
PreviousTaskName_0	SV	E116:EPTTracker:PreviousTaskName
PreviousTaskName_i	SV	E116:EPTTracker:PreviousTaskName.<#>
PreviousTaskType_0	SV	E116:EPTTracker:PreviousTaskType
PreviousTaskType_i	SV	E116:EPTTracker:PreviousTaskType.<#>
TaskName_0	SV	E116:EPTTracker:TaskName
TaskName_i	SV	E116:EPTTracker:TaskName.<#>
TaskType_0	SV	E116:EPTTracker:TaskType
TaskType_i	SV	E116:EPTTracker:TaskType.<#>
TrackerEventID_0	SV	E116:EPTTracker:TrackerEventID
TrackerEventID_i	SV	E116:EPTTracker:TrackerEventID.<#>
Transition_0	SV	E116:EPTTracker:Transition
Transition_i	SV	E116:EPTTracker:Transition.<#>
TransitionTimestamp_0	SV	E116:EPTTracker:TransitionTimestamp
TransitionTimestamp_i	SV	E116:EPTTracker:TransitionTimestamp.<#>

13.3 *Well-Known Names for Collection Events*

13.3.1 This section describes well-known names for collection events currently defined in the Standard.

13.3.2 When collection events correspond to a state model, the well-known name uses the naming convention ‘E116:<state model name>:<transition number>:<from state>-<to state>’ where ‘<state model name>’ is the name of the state model without spaces, ‘<transition number>’ is the respective state model transition number using three digits and leading zeros when necessary, ‘<from state>’ is the name of the initial state without spaces, and ‘<to state>’ is the name of the ending state without spaces. The name ‘NoState’ is used to describe the initial state when created and also the final transition when destroyed. ‘EPTEquipment’ is the state model name for the equipment instance of Equipment Performance Tracking and ‘EPTModule’ is the state model name for the module instance.

Table 9 Well-Known Names for Active Collection Events

Collection Event	*Well-Known Name*
EPTStateChange$_0$	E116:EPTEquipment:EPTStateChange
EPTStateChange$_i$	E116:EPTModule:EPTStateChange.<#>
EPTEquipmentTransition1	When implemented as one collection event: E116:EPTEquipment:00001:NoState-EPTStateModel

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Collection Event	*Well-Known Name*
	When implemented as three collection events: E116:EPTEquipment:001:NoState-Idle E116:EPTEquipment:001:NoState-Busy E116:EPTEquipment:001:NoState-Blocked
EPTEquipmentTransition2	E116:EPTEquipment:002:Idle-Busy
EPTEquipmentTransition3	E116:EPTEquipment:003:Busy-Idle
EPTEquipmentTransition4	E116:EPTEquipment:004:Busy-Busy
EPTEquipmentTransition5	E116:EPTEquipment:005:Busy-Blocked
EPTEquipmentTransition6	E116:EPTEquipment:006:Blocked-Busy
EPTEquipmentTransition7	E116:EPTEquipment:007:Blocked-Idle
EPTEquipmentTransition8	E116:EPTEquipment:008:Idle-Blocked
EPTEquipmentTransition9	E116:EPTEquipment:009:Blocked-Blocked
EPTModuleTransition1	When implemented as one collection event: E116:EPTModule:001:NoState-EPTStateModel When implemented as three collection events: E116:EPTModule:001:NoState-Idle E116:EPTModule:001:NoState-Busy E116:EPTModule:001:NoState-Blocked
EPTModuleTransition2	E116:EPTModule:002:Idle-Busy
EPTModuleTransition3	E116:EPTModule:003:Busy-Idle
EPTModuleTransition4	E116:EPTModule:004:Busy-Busy
EPTModuleTransition5	E116:EPTModule:005:Busy-Blocked
EPTModuleTransition6	E116:EPTModule:006:Blocked-Busy
EPTModuleTransition7	E116:EPTModule:007:Blocked-Idle
EPTModuleTransition8	E116:EPTModule:008:Idle-Blocked
EPTModuleTransition9	E116:EPTModule:009:Blocked-Blocked

13.4 *Well-Known Names for Alarms*

13.4.1 This section describes well-known names for alarms currently defined in the Standard.

13.4.2 This table is deliberately left empty as there are no explicit alarms defined in the Standard.

Table 10 Well-Known Names for Active Alarms

Alarm	*Well-Known Name*

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14 Requirements for Compliance

14.1 Table 11 provides a checklist for EPT compliance.

Table 11 EPT Compliance Statement

Fundamental EPT Requirements	*EPT Section*	*Implemented*	*EPT Compliant*
EPT State Model for Equipment	9	☐ Yes ☐ No	☐ Yes ☐ No
EPT State model for each EPT module	9	☐ Yes ☐ No	☐ Yes ☐ No
EPT Tracker Objects	10	☐ Yes ☐ No	☐ Yes ☐ No
EPT Events	11	☐ Yes ☐ No	☐ Yes ☐ No
EPT Transition-Specific Events	11.4	☐ Yes ☐ No	☐ Yes ☐ No
EPT Tracker Status Variables	12	☐ Yes ☐ No	☐ Yes ☐ No

15 Related Documents

15.1 *Subordinate Standards*

SEMI E116.1 — Specification for SECS-II Protocol for Equipment Performance Tracking (EPT)

15.2 *Other Documents*

International SEMATECH Equipment Performance Management User Requirements Document

ISMT/J300E Equipment Performance Management Operator Requirements Document (URD)

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APPENDIX 1
WELL-KNOWN NAMES FOR RETIRED VARIABLES, COLLECTION EVENTS AND ALARMS

NOTICE: The material in this Appendix is an official part of SEMI E116 and was approved by full letter ballot procedures on December 14, 2023.

A1-1 Background

A1-1.1 This Appendix describes well-known names for items previously defined in the Standard that are not defined in the current revision. They are provided for backwards compatibility.

A1-1.2 For completeness, well-known tables for variables, collection events and alarms included yet deliberately left empty when no applicable variable, collection event, or alarm is defined in this Specification.

A1-1.3 The Last Revision column captures the last revision of the Standard where the item was defined.

A1-2 Well-Known Names for Retired Variables

A1-2.1 This section describes well-known names for variables previously defined in the Standard.

A1-2.2 Refer to SEMI E5 for more information on the variable classes.

Table A1-1 Well-Known Names for Retired Variables

Variable	*Variable Class*	*Well-Known Name*	*Last Revision*

A1-3 Well-Known Names for Retired Collection Events

A1-3.1 This section describes well-known names for collection events previously defined in the Standard.

Table A1-2 Well-Known Names for Retired Collection Events

Collection Event	*Well-Known Name*	*Last Revision*

A1-4 Well-Known Names for Retired Alarms

A1-4.1 This section describes well-known names for alarms previously defined in the Standard.

Table A1-3 Well-Known Names for Retired Alarms

Alarm	*Well-Known Name*	*Last Revision*

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RELATED INFORMATION 1
EXAMPLES OF EQUIPMENT PERFORMANCE TRACKING (EPT)

NOTICE: This Related Information is not an official part of SEMI E116 and was derived from the work of the Information & Control Global Technical Committee. This Related Information was approved for publication by full letter ballot procedures on April 30, 2002.

R1-1 Referenced Standards and Documents

R1-1.1 *SEMI Standards and Safety Guidelines*

SEMI E84 — Specification for Enhanced Carrier Handoff Parallel I/O Interface

SEMI E87 — Specification for Carrier Management (CMS)

SEMI E90 — Specification for Substrate Tracking

SEMI E101 — Guide for EFEM Functional Structure Model

NOTICE: Unless otherwise indicated, all documents cited shall be the latest published versions.

R1-2 Examples of EPT Modules per Tool Type

R1-2.1 Table R1-1 lists examples of EPT modules for various equipment types. It is not required that each tool type support all or be limited to only the modules listed; rather, the listing is provided as an example only.

Table R1-1 Example EPT Modules

Tool Type	*Modules*								
CMP/ Planar	Loadports	Alignment Stage	Wafer Handler	Transport	Polish Table	Scrub Station	Rinse Station		
Linked Litho Track	Loadports	Central Wafer Handler Module	Spin Cup	Link Transfer Station	Buffer	Vapor Prime	Chill Plate	Soft Bake Oven	
Linked Litho Expose	Expose Stage	Wafer Transfer	Link Transfer Station – In port	Link Transfer Station – Out port	Reticle Stage	Reticle Library	Reticle Loader Arm		
Thin Film Sputter	Loadports	Central Wafer Handler(s)	Cool Down	PVD	Airlock	Argon Sputter Etch	Degas	In Situ Metrology	CVD
Diffusion Furnace	Loadports	Quartz Loading Station	Reactor Module	Wafer Transfer System	Buffer				
Implanter	Loadports	Central Wafer Handler(s)	Implant Chamber	Platen	Notch Aligner	Coldplate			
Plater	Loadports	Central Wafer Handler	Degas	Plate Chamber					
Thermal Anneal	Loadports	Central Wafer Handler	Bake Stage						
Wet Etcher	Loadports	Central Wafer Handler	Aligner	Bath CH 1	Bath CH 2	Buffer			

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Tool Type	*Modules*								
Dry Etcher	Loadports	Central Wafer Handler	Aligner	Chill Plate	Etch CH 1	Etch CH 2			
Wafer Sorter	Loadports	Central Wafer Handler	Aligner						
Metrology	Loadports	Central Wafer Handler	Alignment Stage	Measurement Stage					
Inspection	Loadports	Inspection Stage	Handler						
Probe/Test	Loadports	Probe/Test Platform	Handler						

R1-3 EPT Module/Task Definition Example

R1-3.1 Figure R1-1 shows a simple module and the associated actions/tasks that correspond. The EPT Tasks have been diluted to the level that will allow a serial chain of tasks to occur. The Internal Tasks column is present to show that the equipment may have much more happening concurrently at a more detailed level. Table R1-2 shows a possible task-based process flow in this module.

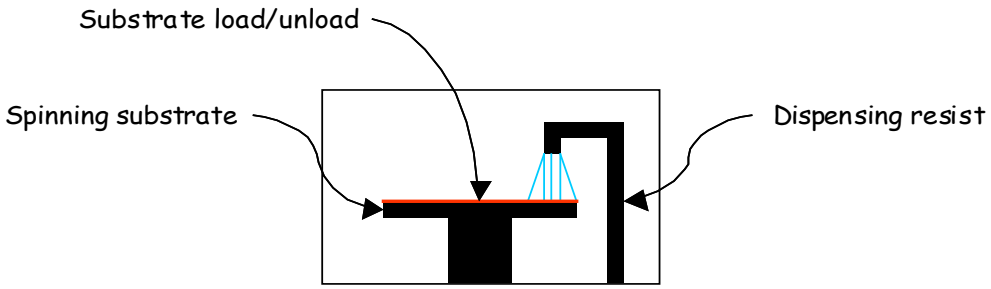

Figure R1-1
Example EPT Module

Table R1-2 Example EPT Tasks

Step	*Action*	*Possible Internal Tasks*	*EPT Active Task*
1	Loading the substrate	Receiving the substrate Substrate alignment	Substrate load
2	Coating the substrate	Substrate spin (accelerate) Substrate spin (steady) Dispense resist Substrate spin (decelerate)	Substrate coat
3	Unloading the substrate	Substrate alignment Substrate removal	Substrate unload

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R1-4 An Example of Equipment Performance Tracking on a Carrier Handling Module

R1-4.1 This section provides examples of equipment performance tracking to clarify the application of the standard to modules that are part of the Equipment Front-End Module (EFEM).

R1-5 Carrier Handling Module (CHM)

R1-5.1 The Carrier Handler is a part of EFEM, excluding any Substrate Handler, which is defined in SEMI E101. A Carrier Handling Module is similar but different from the definition. It is a more general subsystem to handle carriers without handling substrates in EFEM that is applicable to various analyses and applications. The Carrier Handling Module is a logical module that may or may not represent an existing physical module.

R1-5.2 The advantages of applying Equipment Performance Tracking to the Carrier Handling Module are as follows:

- Any equipment, except certain specific ones such as an expose tool within a linked litho tool in the lithography process, has one or more Carrier Handling Modules.

R1-5.2.1 The state of the Carrier Handling Module leads to whole input and output of material to and from an equipment.

- Monitoring the Equipment Performance Tracking state of the Carrier Handling Module helps control loading to or unloading from equipment on appropriate time.
- The state can reveal what related SEMI Standards (i.e., SEMI E87 and SEMI E90) may not address.
- Estimation of capabilities about subcomponents on a Carrier Handling Module is available with tracking the state.
- Each component that makes up the Carrier Handling Module may experience mechanical failure. This can cause the entire Carrier Handling Module to be in an out of service state. Tracking time between failures would be valuable.

R1-6 Components to track performance of Carrier Handling Module

R1-6.1 Generally, a Carrier Handling Module has the following components:

- Carrier Handoff PIO (SEMI E84) Port
- Carrier Transfer Mechanism
- Carrier ID Reader
- Carrier ID Writer
- FOUP Opener
- Carrier Slot Mapper

R1-6.2 This section investigates the following composition to represent examples for Fixed Buffer type and Internal Buffer type equipment. An example of a Carrier Handling Module is illustrated in Figure R1-2. Number of installed components are listed in the Table R1-3.

Table R1-3 Typical Composition of Carrier Handoff Module

Number Installed	*Carrier Handoff PIO*	*Carrier Transfer Mechanism*	*Carrier ID Reader*	*Carrier ID Writer*	*FOUP Opener*	*Slot Mapper*
Internal Buffer Type	2	4	2	1	1	1
Fixed Buffer Type	2	2	2	2	2	2

#1 Several Carrier Transfer Mechanisms are usually docking mechanism for Fixed Buffer type. For Internal Buffer type, one of them is a Carrier Transfer Robot in internal buffer, one is a Docking Mechanism inside the internal buffer and the other one is Carrier Transfer Mechanism to pass carrier between Loadport and the Internal Buffer.

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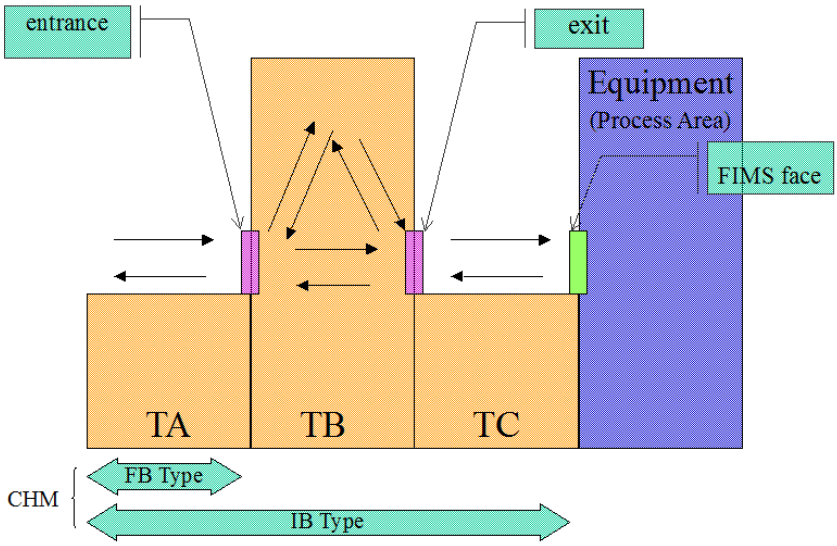

Figure R1-2
Carrier Handling Module

R1-6.3 In Figure R1-2, the following translation applies:

- FB Type = Fixed Buffer Type
- IB Type = Internal Buffer Type
- TA = Transport A
- TB = Transport B
- TC = Transport C

R1-7 Triggers of Transition for each Components of Carrier Handling Module

R1-7.1 The following tables describe triggers to change the state of components for specific tasks.

Table R1-4 Triggers of Transitions for Specific Tasks of Carrier Handoff PIO

#	*Previous State*	*Trigger*	*New State*	*Action(s)*	*Comment*
2	IDLE	Receiving VALID signal in SEMI E84.	BUSY	Set off handoff sequence	Loading or Unloading sequence. Processing or moving wafers between carrier and process container before Unloading.
3	BUSY	Receiving COMPT signal in SEMI E84 or Timeout in handoff sequence.	IDLE	Waiting disconnected	Loading or Unloading sequence is completed or ended. Processing or moving wafers between carrier and process container after Loading.

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Table R1-5 Triggers of Transitions for Specific Tasks of Carrier Transfer Mechanism on Loadport of Fixed Buffer Type or FIMS Port of IB

#	*Previous State*	*Trigger*	*New State*	*Action(s)*	*Comment*
2	IDLE	Docking or Undocking Started	BUSY	None	Carrier moving sequence before docking. Processing or moving wafers between carrier and process container before undocking.
3	BUSY	Docking or Undocking Complete	IDLE	None	Processing or moving wafers between carrier and process container after docking.

Table R1-6 Triggers of Transitions for Specific Tasks of Carrier Transfer Mechanism in Internal Buffer Unit

#	*Previous State*	*Trigger*	*New State*	*Action(s)*	*Comment*
2	IDLE	Transfer Started	BUSY	Moves carrier to destination	
3	BUSY	Transfer Complete	IDLE	None	Ready to unload or remove from port.

Table R1-7 Triggers of Transitions for Specific Tasks of Carrier ID Reader

#	*Previous State*	*Trigger*	*New State*	*Action(s)*	*Comment*
2	IDLE	Reading Started	BUSY	None	Carrier moving sequence before reading.
3	BUSY	Reading Complete	IDLE	Verification	Processing or moving wafers between carrier and process container after opening.

Table R1-8 Triggers of Transitions for Specific Tasks of Carrier ID Writer

#	*Previous State*	*Trigger*	*New State*	*Action(s)*	*Comment*
2	IDLE	Writing Started	BUSY	None	Processing wafers before writing.
3	BUSY	Writing Complete	IDLE	Waiting next writing or any post writing action	Carrier moving sequence after writing.

Table R1-9 Triggers of Transitions for Specific Tasks of FOUP Opener

#	*Previous State*	*Trigger*	*New State*	*Action(s)*	*Comment*
2	IDLE	Opening or Closing Started	BUSY	None	Carrier moving sequence before opening.
3	BUSY	Opening or Closing Complete	IDLE	None	Processing or moving wafers between carrier and process container after opening.

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Table R1-10 Triggers of Transitions for Specific Tasks of Slot Mapper

#	*Previous State*	*Trigger*	*New State*	*Action(s)*	*Comment*
2	IDLE	Scanning Started	BUSY	None	Carrier moving sequence before mapping.
3	BUSY	Scanning Complete	IDLE	Waiting next scan	Processing or moving wafers between carrier and process container after mapping.

Table R1-11 Triggers of Transitions Common for all Components of Carrier Handling Module

#	*Previous State*	*Trigger*	*New State*	*Action(s)*	*Comment*
1	(no state)	Initialization is complete	IDLE	None	
7	BLOCKED	Get Recovered	IDLE	None	
8	IDLE	Unable to do task	BLOCKED	None	
9	BLOCKED	Get another Fault	BLOCKED	None	

R1-8 Operation Scenarios for Carrier Handling Module

R1-8.1 A couple of following scenarios are typical for different type of Carrier Handling Module.

R1-8.2 *Carrier Handling Module for Fixed Buffer Type*

R1-8.2.1 Beside the number of Loadports, following table shows just one set of components out of them.

R1-8.2.2 Indicated subsystems: PIO = Carrier Handoff PIO Port, CID-R = Carrier ID Reader, X'fer Transfer Mechanism = Carrier Transfer Mechanism (Docking Mechanism), CID-W = Carrier ID Writer, Opener = FOUP Opener, Mapper = Carrier Slot Mapper.

Table R1-12 Scenario of Fixed Type Carrier Handling Module

Event #	*Description*	*Time*	*PIO*	*CID-R.*	*X'fer Mech*	*CID-W*	*Opener*	*Mapper*
0	Carrier Handling Module and all components starting in 'IDLE'	0:00	IDLE	IDLE	IDLE	IDLE	IDLE	IDLE
1	Loading Started	0:30	BUSY	↑	↑	↑	↑	↑
2	Loading Completed	1:00	IDLE	↑	↑	↑	↑	↑
3	ID Reading Started	1:05	↑	BUSY	↑	↑	↑	↑
4	ID Reading Complete	1:10	↑	IDLE	↑	↑	↑	↑
5	ID Verification	-	↑	↑	↑	↑	↑	↑
6	Docking Started	1:20	↑	↑	BUSY	↑	↑	↑
7	Docking Complete	1:50	↑	↑	IDLE	↑	↑	↑
8	Opening Start	1:55	↑	↑	↑	↑	BUSY	↑
9	Opening Complete	2:25	↑	↑	↑	↑	IDLE	↑
10	Mapping Started	2:30	↑	↑	↑	↑	↑	BUSY
11	Mapping Complete	2:15	↑	↑	↑	↑	↑	IDLE
12	Slotmap Verification	-	↑	↑	↑	↑	↑	↑
13	Process Started	12:15	↑	↑	↑	↑	↑	↑
14	Process Completed	47:15	↑	↑	↑	↑	↑	↑
15	Writing Started	47:20	↑	↑	↑	BUSY	↑	↑
16	Writing Complete	48:10	↑	↑	↑	IDLE	↑	↑

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Event #	*Description*	*Time*	*PIO*	*CID-R.*	*X'fer Mech*	*CID-W*	*Opener*	*Mapper*
17	Closing Start	48:10	↑	↑	↑	↑	BUSY	↑
18	Closing Complete	48:40	↑	↑	↑	↑	IDLE	↑
19	Undocking Started	48:45	↑	↑	BUSY	↑	↑	↑
20	Undocking Complete	49:15	↑	↑	IDLE	↑	↑	↑
21	Unloading Started	52:25	BUSY	↑	↑	↑	↑	↑
22	Unloading Complete	52:55	IDLE	↑	↑	↑	↑	↑
23	Carrier Handling Module and all components returning in 'IDLE'	52:55	↑	↑	↑	↑	↑	↑

R1-8.3 *Carrier Handling Module for Internal Buffer Type*

R1-8.3.1 It is assumed that Carrier Slot Mapper and Carrier ID Writer are mounted at an FIMS port next to Internal Buffer located inside of the equipment. The following table shows the set of components of a single Loadport.

R1-8.3.2 Indicated subsystems: PIO = Carrier Handoff PIO Port, CID-R = Carrier ID Reader, X'fer A = Carrier Transfer Mechanism from Handoff position to Carrier In/Out position, X'fer B = Carrier Transfer Mechanism of Internal Buffer, X'fer C = Carrier Transfer Mechanism (Docking Mechanism) for internal FIMS port, CID-W = Carrier ID Writer, Opener = FOUP Opener, Map = Carrier Slot Mapper.

Table R1-13 Scenario of Internal Buffer Type Carrier Handling Module

Event #	*Description*	*Time*	*PIO*	*CID-R.*	*X'fer A*	*X'fer B*	*X'fer C*	*CID-W*	*Opener*	*Map*
0	Carrier Handling Module and all components starting in 'IDLE'	0:00	IDLE	IDLE	IDLE	IDLE	IDLE	IDLE	IDLE	IDLE
1	Loading Started	0:30	BUSY	↑	↑	↑	↑	↑	↑	↑
2	Loading Completed	1:00	IDLE	↑	↑	↑	↑	↑	↑	↑
3	ID Reading Started	1:05	↑	BUSY	↑	↑	↑	↑	↑	↑
4	ID Reading Complete	1:10	↑	IDLE	↑	↑	↑	↑	↑	↑
5	ID Verification	-	↑	↑	↑	↑	↑	↑	↑	↑
6	Transfer Started	1:20	↑	↑	BUSY	↑	↑	↑	↑	↑
7	Transfer Complete	1:35	↑	↑	IDLE	↑	↑	↑	↑	↑
8	Moving Started	1:40	↑	↑	↑	BUSY	↑	↑	↑	↑
9	Moving Complete	2:25	↑	↑	↑	IDLE	↑	↑	↑	↑
10	Docking Started	2:30	↑	↑	↑	↑	BUSY	↑	↑	↑
11	Docking Complete	2:50	↑	↑	↑	↑	IDLE	↑	↑	↑
12	Opening Start	2:55	↑	↑	↑	↑	↑	↑	BUSY	↑
13	Opening Complete	3:25	↑	↑	↑	↑	↑	↑	IDLE	↑
14	Mapping Started	3:30	↑	↑	↑	↑	↑	↑	↑	BUSY
15	Mapping Complete	3:15	↑	↑	↑	↑	↑	↑	↑	IDLE
16	Slotmap Verification	-	↑	↑	↑	↑	↑	↑	↑	↑
17	Wafer Moving Started	3:30	↑	↑	↑	↑	↑	↑	↑	↑
18	Wafer Moving Completed	7:00	↑	↑	↑	↑	↑	↑	↑	↑
12	Closing Start	7:05	↑	↑	↑	↑	↑	↑	BUSY	↑
13	Closing Complete	7:20	↑	↑	↑	↑	↑	↑	IDLE	↑
21	Undocking Started	7:25	↑	↑	↑	↑	BUSY	↑	↑	↑
22	Undocking Complete	7:45	↑	↑	↑	↑	IDLE	↑	↑	↑
8	Moving Started	7:50	↑	↑	↑	BUSY	↑	↑	↑	↑

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Event #	*Description*	*Time*	*PIO*	*CID-R.*	*X'fer A*	*X'fer B*	*X'fer C*	*CID-W*	*Opener*	*Map*
9	Moving Complete	8:35	↑	↑	↑	IDLE	↑	↑	↑	↑
17	Process Started	16:15	↑	↑	↑	↑	↑	↑	↑	↑
18	Process Completed	147:15	↑	↑	↑	↑	↑	↑	↑	↑
8	Moving Started	146:30	↑	↑	↑	BUSY	↑	↑	↑	↑
9	Moving Complete	147:15	↑	↑	↑	IDLE	↑	↑	↑	↑
10	Docking Started	147:20	↑	↑	↑	↑	BUSY	↑	↑	↑
11	Docking Complete	147:40	↑	↑	↑	↑	IDLE	↑	↑	↑
19	Writing Started	147:45	↑	↑	↑	↑	↑	BUSY	↑	↑
20	Writing Complete	148:10	↑	↑	↑	↑	↑	IDLE	↑	↑
12	Opening Start	148:10	↑	↑	↑	↑	↑	↑	BUSY	↑
13	Opening Complete	148:25	↑	↑	↑	↑	↑	↑	IDLE	↑
17	Wafer Moving Started	148:30	↑	↑	↑	↑	↑	↑	↑	↑
18	Wafer Moving Completed	152:00	↑	↑	↑	↑	↑	↑	↑	↑
12	Closing Start	152:05	↑	↑	↑	↑	↑	↑	BUSY	↑
13	Closing Complete	152:20	↑	↑	↑	↑	↑	↑	IDLE	↑
21	Undocking Started	152:25	↑	↑	↑	↑	BUSY	↑	↑	↑
22	Undocking Complete	152:45	↑	↑	↑	↑	IDLE	↑	↑	↑
23	Moving Started	152:50	↑	↑	↑	BUSY	↑	↑	↑	↑
24	Moving Complete	153:35	↑	↑	↑	IDLE	↑	↑	↑	↑
25	Transferring Started	153:40	↑	↑	BUSY	↑	↑	↑	↑	↑
26	Transferring Complete	153:55	↑	↑	IDLE	↑	↑	↑	↑	↑
27	Unloading Started	202:25	BUSY	↑	↑	↑	↑	↑	↑	↑
28	Unloading Complete	202:55	IDLE	↑	↑	↑	↑	↑	↑	↑
29	Carrier Handling Module and all components returning in 'IDLE'	202:55	↑	↑	↑	↑	↑	↑	↑	↑

R1-9 Tracking Performance for Whole Carrier Handling Module

R1-9.1 Even composition and property of equipment are dependent on equipment type, operation and usage depends on equipment user. Because actual usage deeply affects the whole performance of the Carrier Handling Module and/or Equipment, the user is responsible for reconstructing the performance from performance tracking events for each component.

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